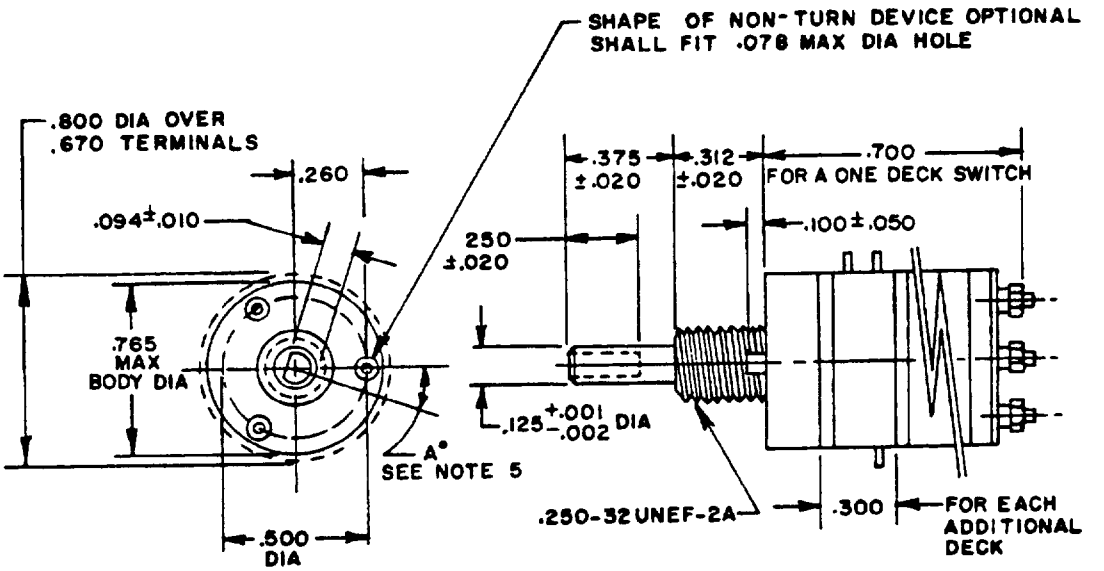
MILITARY SPECIFICATION SHEET

SWITCHES, ROTARY, CLOSED CONSTRUCTION, 0 5 AMPERE,
STYLE SR

This specification is approved for use by Engineering and Technical Support Division (AFLC/LOIE), Department of the Air Force, and is available for use by All Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switches described herein shall consist of this document and the latest issue of Specification MIL-S-3786.



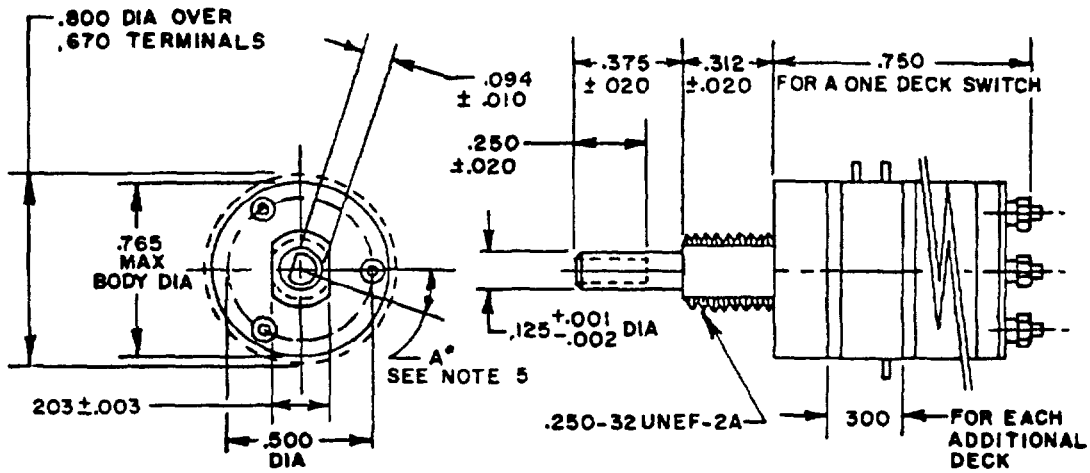
"E" CONSTRUCTION STYLE
(EXPLOSION-PROOF, CLOSED CONSTRUCTION WITHOUT SEALED SHAFT AND BUSHING)

INCHES	MM	INCHES	MM	INCHES	MM
001	03	094	2.39	312	7 92
002	05	100	2 54	375	9 53
010	25	125	3 18	500	12 70
020	51	250	6 35	670	17 02
050	1 27	260	6.60	700	17 78
078	1 98	300	7 62	765	19 43
				800	20 32

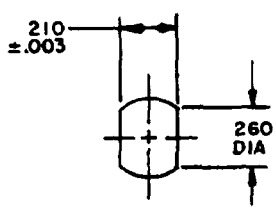
NOTES
1 Dimensions are in inches.
2 Metric equivalents (to the nearest 01 mm) are given for general information only and are based upon 1 inch = 25 4 mm
3 Unless otherwise specified, tolerances are ± 015 (38 mm) and ±3° on angles (non-accumulative)
4 Maximum weight one deck switch is 16 grams Add 2 5 max grams for each additional deck and based upon 1 oz = 28 35 grams
5 Shaft flat angle A° is the angle between a line through the center of the shaft, and center of the nonturn device and another line through the center of the shaft and perpendicular to the shaft flat, with switch in position number 1 This refers to panel position, see figure 2 for terminal location

FIGURE 1 Style SR41 switch.

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"F" CONSTRUCTION STYLE
(EXPLOSION-PROOF, CLOSED CONSTRUCTION WITH SEALED SHAFT AND BUSHING)



MOUNTING HOLE

INCHES	MM	INCHES	MM	INCHES	MM
001	03	125	3 18	312	7 92
002	05	203	5 16	375	9 53
003	08	210	5 33	500	12 70
010	25	250	6 35	670	17 02
020	51	260	6 60	750	19 05
094	2 39	300	7 62	765	19 43
				.800	20 32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25 4 mm
3. Unless otherwise specified, tolerances are ± 015(38 mm) and ±3° on angles (non-accumulative).
4. Maximum weight: one deck switch is 16 grams. Add 2.5 max grams for each additional deck and based upon 1 oz = 28.35 grams
5 Shaft flat angle A° is the angle between a line through the center of the shaft, perpendicular to the mounting bushing flats and another line through the center of the shaft and perpendicular to the shaft flat, with switch in position number 1. This refers to panel position; see figure 2 for terminal location

FIGURE 1 Style SR41 switch - Continued

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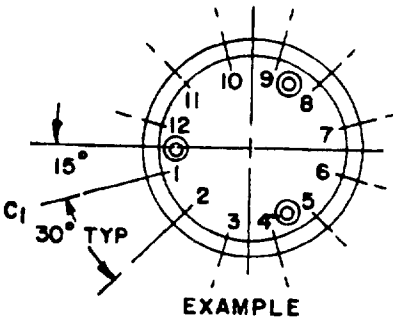


FIGURE 2 Rear view of terminal configurations

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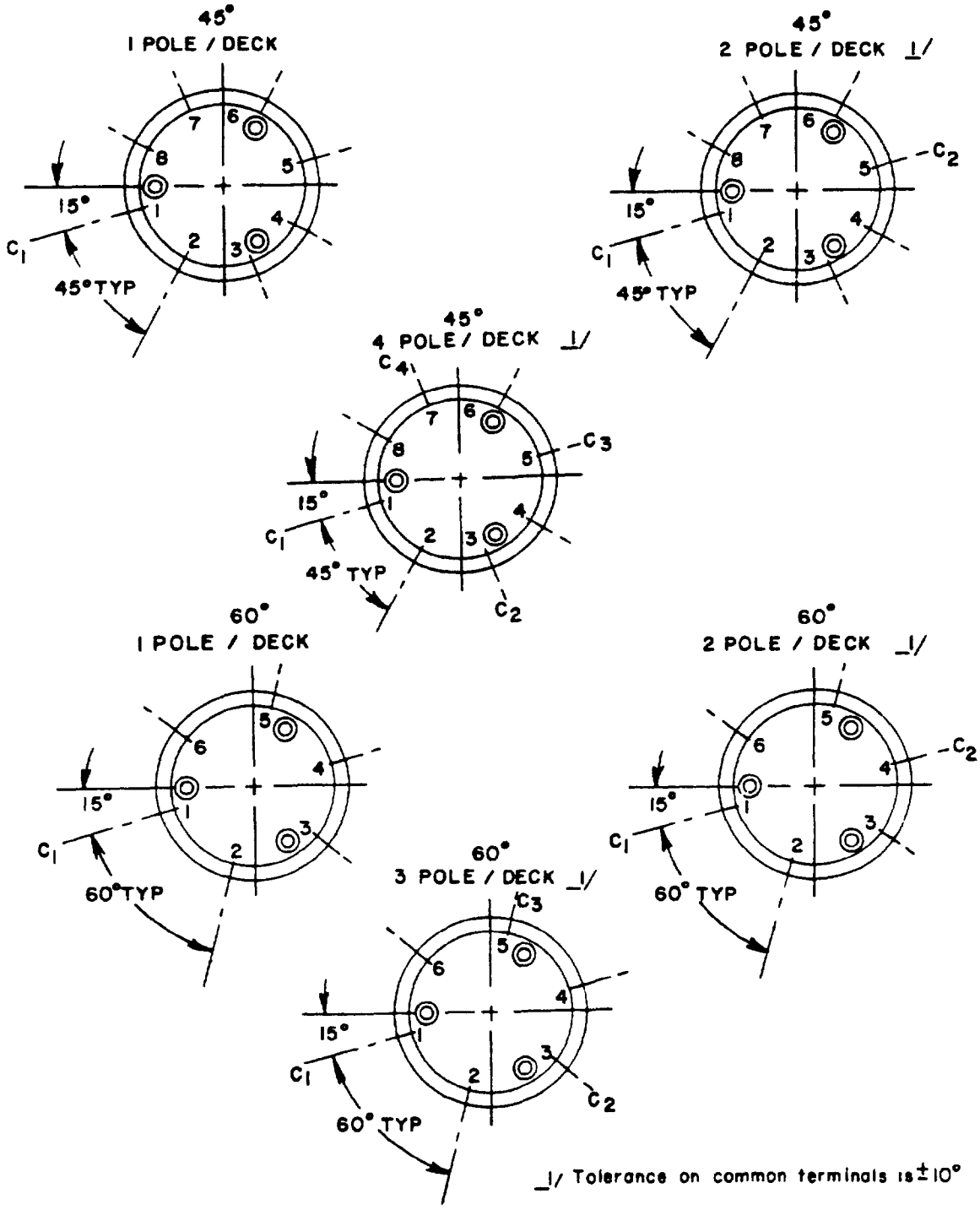


FIGURE 2 Rear view of terminal configurations - Continued

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REQUIREMENTS

Dimensions and configurations See figures 1 and 2 Angle "A" is 15° for switches of 30°, 45°, and 60° angle of throw, 18° for switches of 36° angle of throw

Angle of throw and number of poles per deck See table IV.

Insulation material Symbol P.

Construction style See table IV

Temperature-life characteristics· See table V

Number of positions See table V

Vibration grade Symbol 3 (10 to 2,000 Hz)

Shock type Symbol H (high impact) and
M (medium impact).

Altitude Symbol C (70,000 ft)

Number of decks See tables VI and VII.

Mounting hardware Each switch shall be provided with two hexagon nuts, unless otherwise specified, 0.312 ±0.010 inch across flats and 0.078 ±0.010-inch thick, one internal tooth lockwasher .415 maximum OD

Terminals Terminals shall be designed so that the terminals can accomodate a wire having a diameter of 0.030 inch The terminals shall project in a direction parallel to the plane of the switch section

Moisture resistance After conclusion of test, insulation resistance shall be While switches are in humidity chamber - greater than 0.5 megohm At end of drying period - Not less than 10 megohms

Stop strength 10 pound-inches minimum.

Rotational torque The minimum and maximum values of torque determined for shaft rotation shall be within the limits specified in table I

Terminal strength (pull) A force of 2 pounds minimum shall be applied to the terminals

TABLE I Rotation-torque limits

Total number of switch poles	Temperature	Torque (lb. -in)	
		Minimum	Maximum
Less than 4	Room Minimum	.32 .32	1.7 2.2
4 thru 8	Room Minimum	4 4	2.7 3.2
More than 8	Room Minimum	5 5	3.5 4.0

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REQUIREMENTS -Continued

Dielectric withstanding voltage See table II

TABLE II Dielectric-test voltages.

Altitude	Test voltage
	(Volts, rms) (60 Hz)
At atmospheric pressure	500
At reduced barometric pressure	350

Extent of qualification "F" construction style qualifies "E" construction style for the same switches provided two sample units of "E" construction switches are tested to explosion requirement to verify explosion proof capability

Life (rotational) The test loads for the applicable circuit conditions shall be as specified in table III Each of the loads specified for the applicable environmental condition shall be switched by at least one rotor contact of the switch

TABLE III Circuit values for life (rotational) test

Environmental condition	Inductive load (2 8 henries)		Resistive load	
	Milliamperes	Volts, dc	Amperes	Volts
At atmospheric pressure and elevated temperature (85° or 125°C) (25,000 cycles)	50	28	.125 075	28 dc 115 Vrms 60 Hz
At reduced barometric pressure (approx 25°C) (25,000 cycles)	10	28	050 020	28 dc 115 Vrms 60 Hz
At atmospheric pressure and room temperature (approx 25°C) (10,000 cycles) 1/	250	28	500 500	28 dc 115 Vrms 60 Hz

1/ The contact resistance (applicable to ac and dc), after life tests shall not exceed 50 milliohms

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PART NUMBER The part number shall consist of the prefix "M3786/41-" and a dash number consisting of four characters, one each from tables IV, V, VI, and VII as appropriate

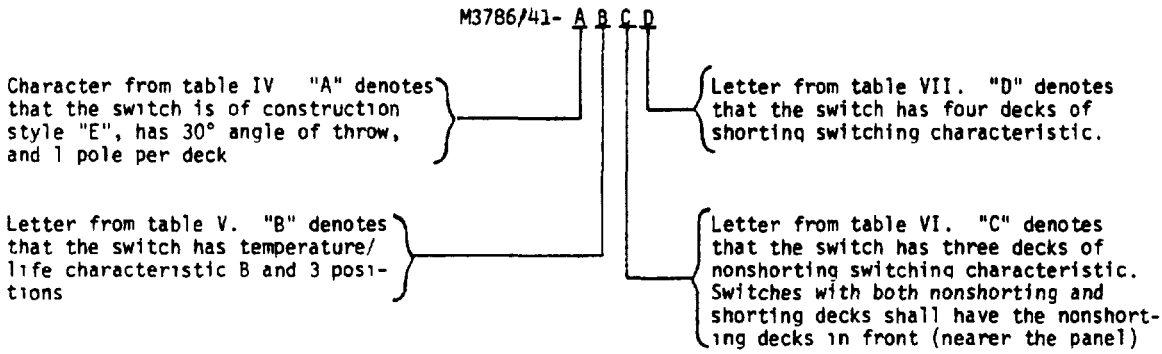


TABLE IV.

Code character for combination of construction style, angle of throw, and number of poles per deck

Code letter	Construction style	Angle of throw	Poles per deck
A	E	30	1
B	E	30	2
C	E	30	3
D	E	30	4
E	E	30	6
F	E	36	1
G	E	36	2
H	E	45	1
J	E	45	2
K	E	45	4
L	E	60	1
M	E	60	2
N	E	60	3
P	F	30	1
Q	F	30	2
R	F	30	3
S	F	30	4
T	F	30	6
U	F	36	1
V	F	36	2
W	F	45	1
X	F	45	2
Y	F	45	4
Z	F	60	1
8	F	60	2
9	F	60	3

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TABLE V.

Code letter for combination

of temperature-life characteristic

and number of switch positions

Code letter	Temperature-life characteristic	Number of positions
A	B	2
B	B	3
C	B	4
D	B	5
E	B	6
F	B	7
G	B	8
H	B	9
J	B	10
K	B	11
L	B	12
M	B	C 1/
N	C	2
P	C	3
Q	C	4
R	C	5
S	C	6
T	C	7
U	C	8
V	C	9
W	C	10
X	C	11
Y	C	12
Z	C	C 1/

1/ Switch is continuous rotation type (no stops), and number of positions is dependent on angle of throw, 30° - 12 pos , 36° - 10 pos , 45° - 8 pos , and 60° - 6 positions.

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TABLE VI

Code letter for number of decks with

Nonshorting switching characteristics

Code letter	Number of decks with nonshorting switching
A	0
B	1
C	2
D	3
E	4
F	5
G	6
H	7
J	8
K	9
L	10

TABLE VII.

Code letter for number of decks with

shorting switching characteristics

Code letter	Number of decks with shorting switching
A	0
B	1
C	2
D	3
E	4
F	5
G	6
H	7
J	8
K	9
L	10

Custodian.
Air Force - 85

Review activity:
[illegible] - 99

Preparing activity:
Air Force - 85

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MIL-S-3786/41 (USAF) Switches, Rotary, Closed Construction

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER FSC 5930

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